# Computer Architecture
# Lab (Week 4)

# Assembly Directive

- ❑ What are assembler directives?
- ❑ How do you set up an area of memory to be data?
  - For temporary data of the program
  - Set up constants for use in my program
  - And other possible uses
- ➢ Assembler Directives are a means by which you can 'direct' the assembler to take specific actions.

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# Assembly Directive

| Assembly Directive | Syntax | Example | Description |
|---|---|---|---|
| @NEXT | @NEXT Memory_Address | @NEXT 4000 | This assembler directive specifies where the next instruction is to be assembled in memory. |
| @BEGIN | @BEGIN Memory_Address | @BEGIN 2000 | - Specify the memory location for execution begin<br>- Can start anywhere in the program.<br>- If there multi @BEGIN is used only the last one is considered. |
| @EQU | Label: @EQU Value | numA: @EQU 01 | This assembler directive is used to declare constants. The symbol name is<br>given as a label. |

# TP01: Calculate the sum of series of 8 bits numbers (no carry)

- Statement: Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H.
- Sample Problem: 2000H = 05, 2001H = 04H

  2002H = 50, 2003H = 10H
- Result: (2005) = 69H

# TPO2: Calculate the sum of series of 8 bits numbers (with carry)

- Statement: Calculate the sum of series of 8-bits data from memory location 2000H to 2003H using branching instruction to loop. Memory location 2010H will be store the amount of series number to sum. Then store the sum data to address 2005H.
- Sample Problem: 2000H = 5A, 2001H = 10H
  2002H = 74, 2003H = 85H
- Result: (2005) = 63H, (2006) = 01H

# TP03: Calculate the sum of 1+2+…+n

- Statement: Calculate the sum from 1 to n number in memory location 2000H. Then store to address 2001H.
- Sample Problem: (2000H) = 04H
- Result: (2001H) = 01 + 02 + 03 + 04 = 0A

TP04: Write an assembly language to calculate square of a number.

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# TP05: Write an assembly language to calculate (a+b)2 and store answer in location (2005)

# TP06: Write an assembly language to calculate whether m X 2n and store answer in address 2000

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# TP07: Write an assembly language to calculate (a-b)2

# TP08: Write an assembly language to calculate whether (a+b)2 or (a-b)2

# Thanks